

CSCE 647 IMAGE SYNTHESIS · MODULE 01 · LECTURE NOTES

Radiometry: What Rendering Computes

Five quantities, one of which is the one you actually want.

MODULE OUTCOMES

Define flux, irradiance, intensity, and radiance, and say which one a pixel holds.

Use solid angle and the projected-area cosine correctly.

Explain why radiance is invariant along a ray in a vacuum, and why that makes ray tracing possible.

State the BRDF as a ratio of differentials and give its three required properties.

Explain what a pixel value is an integral of.

1 Why a whole module on units

A renderer that confuses irradiance with radiance produces images that are wrong by a factor involving π and a cosine, and such images look entirely plausible. The error does not announce itself — it shows up as 'my renderer is darker than the reference and I do not know why', which is the most common complaint in this subject.

There are five quantities. They are the same physical thing divided by different amounts.

Quantity	Symbol	SI units	Definition
Radiant energy	Q	joule (J)	Total energy. Rarely used directly in rendering.
Radiant flux, or power	Φ	watt (W)	dQ/dt . Energy per unit time. A light bulb's output.
Irradiance	E	W/m^2	$d\Phi/dA$. Power arriving per unit surface area, summed over all incoming directions.
Radiant intensity	I	W/sr	$d\Phi/d\omega$. Power per unit solid angle. Natural for point lights, which have no area.
Radiance	L	$W/(m^2 \cdot sr)$	$d^2\Phi / (d\omega dA \cos\theta)$. Power per unit projected area per unit solid angle. The quantity rendering transports.

RADIANCE IS THE QUANTITY YOU WANT, FOR FOUR REASONS

It is complete. Every other radiometric quantity is an integral of radiance, so a renderer that computes radiance has computed everything else implicitly.

It is directional. Radiance is defined per solid angle, so it can describe what a single ray carries. Irradiance has already summed over all incoming directions and destroyed that information.

It is invariant along a ray in a vacuum, developed in §3. This is what makes a ray a lookup rather than an integral.

It is what a sensor responds to. A pixel value is an integral of radiance over the sensor's footprint, aperture, exposure, and spectral response, and nothing else enters.

2 Solid angle

Integrating over directions requires a measure on the set of directions, and that measure is solid angle. The construction is exactly parallel to area: just as an angle in the plane is arc length on the unit circle, a solid angle is area on the unit sphere.

$$d\omega = dA \cos\theta / r^2$$

A patch of area dA at distance r , tilted so its normal makes angle θ with the direction to the viewer, subtends this solid angle. Both factors are doing necessary work: the $\cos\theta$ foreshortens the patch to the area actually presented, and the $1/r^2$ accounts for its apparent size shrinking with distance.

The full sphere has measure 4π steradians and a hemisphere 2π . In spherical coordinates, with θ measured from the surface normal:

$$d\omega = \sin\theta \, d\theta \, d\phi$$

THE SIN θ IS THE MOST COMMONLY DROPPED TERM IN RENDERING

A naive hemisphere integrator that loops over θ and ϕ uniformly and forgets the $\sin\theta$ is integrating against the wrong measure. It over-weights directions near the pole and under-weights those near the horizon.

The symptom is subtle: images look nearly right but are consistently off by a factor, and the error changes with surface orientation, so it cannot be corrected by scaling the output.

The check: integrate the constant function 1 over your hemisphere. You must get 2π . If you get something else, nothing downstream of that integrator can be trusted.

Write that check as a unit test in Module 05 and keep it forever. It costs three lines and catches an entire class of error.

2.1 The cosine, and what it is not

A cosine appears in almost every equation in this course, and it is the same cosine every time: the angle between a direction and the surface normal.

LAMBERT'S COSINE LAW IS GEOMETRY

A beam of light with a fixed cross-section striking a surface head-on illuminates an area A . The same beam striking at angle θ illuminates an area $A/\cos\theta$ — a larger patch. The same power is spread over more surface, so the power *per unit area* is reduced by $\cos\theta$.

That is the entire content of the cosine law. It is not an approximation, not a property of any particular material, and not something a shading model chooses to obey.

The cosine appears whenever you convert between 'per unit area measured on the surface' and 'per unit area measured perpendicular to the beam'.

A Lambertian surface is therefore not 'one that obeys the cosine law', which is how it is usually described and is wrong. Everything obeys the cosine law. A Lambertian surface is one whose *BRDF* is constant in direction — it scatters incoming light equally into all outgoing directions.

3 Invariance along a ray

This is the property the whole algorithm rests on, and it is worth the two minutes it takes to see why it holds.

Consider two small patches, dA_1 and dA_2 , facing each other at distance r , with normals making angles θ_1 and θ_2 with the line between them. The flux leaving the first and arriving at the second is

$$d^2\Phi = L \, dA_1 \cos\theta_1 \, d\omega_1 \quad \text{where} \quad d\omega_1 = dA_2 \cos\theta_2 / r^2$$

This expression is symmetric in the two patches: substituting gives $d^2\Phi = L \, dA_1 \cos\theta_1 \, dA_2 \cos\theta_2 / r^2$, which is unchanged if the labels are swapped. The same flux is leaving the first patch as is arriving at the second, and both are described by the same radiance L . Nothing has been lost in between.

WHY IRRADIANCE FALLS OFF AND RADIANCE DOES NOT

Move a surface twice as far from a light. The irradiance it receives drops by a factor of four — the inverse square law, which is real.

But the solid angle the *light* subtends at the surface also drops by a factor of four. Radiance is power per unit area *per unit solid angle*, so the two factors of four cancel exactly and the radiance is unchanged.

The everyday version: a white wall does not get darker as you walk away from it. Each patch of wall sends you less power, and each patch occupies proportionally less of your visual field. The brightness per unit of visual field — which is radiance, and is what you perceive — is constant.

This also explains why the sun and a photograph of the sun can have the same apparent brightness while differing enormously in total power.

WHAT THIS LICENSES

If radiance were attenuated along a ray, a renderer would have to integrate along every segment of every path through empty space, and the cost of a ray would depend on its length.

Because radiance is invariant, **a ray is a lookup**: trace until you hit a surface, evaluate the radiance leaving that surface toward you, and that value is exactly the radiance arriving at the ray's origin. No integration along the way.

So ray tracing is not an approximation to light transport. It is an exact consequence of radiance invariance in a vacuum, and the approximations in a ray tracer are all in how the integrals at *surfaces* are estimated.

The vacuum assumption is load-bearing. Module 10 fills space with participating media, invariance fails, and the lookup becomes an integral along the ray again — which is why volumetric rendering is so much more expensive than it looks.

4 The BRDF

Having established what travels along a ray, we need the function that describes what happens when it meets a surface.

$$f(p, \omega_i, \omega_o) = dL_o(p, \omega_o) / dE(p, \omega_i) = dL_o(p, \omega_o) / (L_i(p, \omega_i) \cos\theta_i d\omega_i)$$

In words: of the differential irradiance arriving at p from direction ω_i , what fraction is scattered into direction ω_o per unit solid angle? Its units are inverse steradians.

A BRDF IS A DENSITY, SO IT IS NOT BOUNDED BY 1

Students reasonably expect a 'fraction of light reflected' to lie between 0 and 1. The BRDF is not that fraction — it is a *density* with respect to solid angle, and densities can be arbitrarily large where they are concentrated.

A perfect mirror is the extreme case: it reflects all the incoming light into a single direction of zero solid angle, so its BRDF is a Dirac delta and is unbounded. This is not a pathology to be regularised away; it is why specular surfaces need separate handling throughout a renderer (Module 07).

What is bounded is the integral. Energy conservation constrains $\int f \cos\theta d\omega \leq 1$, not f itself.

Property	Statement	Consequence of violating it
Non-negativity	$f \geq 0$ for all directions.	Negative radiance, which propagates and produces black or NaN pixels far from the offending surface.
Reciprocity	$f(p, \omega_i, \omega_o) = f(p, \omega_o, \omega_i)$.	Bidirectional methods give wrong answers — see below.
Energy conservation	$\int f(p, \omega_i, \omega_o) \cos\theta_i d\omega_i \leq 1$ for every ω_o .	Paths gain energy at each bounce. Russian roulette fails to terminate, images brighten without converging, and the renderer may diverge rather than merely look wrong.

RECIPROCITY IS NOT A FORMALITY

Reciprocity states that the BRDF is unchanged when the incoming and outgoing directions are exchanged: light does not care which way along the path it is travelling.

Every bidirectional method in Module 11 depends on it. Light tracing, bidirectional path tracing, and photon mapping all construct paths starting from a light source and then use them as though they had been constructed from the camera. That substitution is valid precisely because the BRDF is reciprocal.

A non-reciprocal BRDF gives different answers depending on how a path happened to be built, which shows up as an inconsistency between integrators on the same scene — an extremely confusing bug, because each integrator is individually self-consistent.

Several widely used real-time shading models are not reciprocal, having never been required to be. A model that served you well in CSCE 641 may be unusable here, and checking is cheap: evaluate it with the arguments swapped.

5 What a pixel actually is

$$\mathbf{I} = \iiint w(\mathbf{x}, \mathbf{y}) \mathbf{L}(\mathbf{x}, \mathbf{y}, \omega, t, \lambda) \mathbf{S}(\lambda) d\lambda dt d\omega d\mathbf{x} d\mathbf{y}$$

Dimension	Integrates over	Which produces
$\mathbf{x}, \mathbf{y}$	The pixel's spatial footprint, weighted by the reconstruction filter w .	Antialiasing
ω	The directions admitted by the lens aperture.	Depth of field
t	The shutter interval.	Motion blur
λ	Wavelength, weighted by the sensor response S .	Colour , and dispersion (Module 13)

FOUR FEATURES, ONE INTEGRAL

In a rasteriser, antialiasing, depth of field, and motion blur are three separate systems with three separate implementations, three sets of artefacts, and three sets of tuning parameters.

In a Monte Carlo renderer they are **three more dimensions of the same integral**, and the machinery that handles one handles all of them without modification: jitter the sample position within the pixel, the sample point on the lens, the sample time within the shutter interval, and the sampled wavelength.

This is the structural advantage of the approach, and it is why offline renderers get these effects almost free while real-time renderers fight for each one separately.

It is also why Module 06 — how to choose those sample points well — pays off across every feature at once.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
PBRT 4th ed. — Chapter 4, Radiometry, Spectra, and Colour	The definitive treatment of this module. §4.1 for the quantities and §4.2 for radiance invariance. Skip the colour sections until Module 13.
Scratchapixel — Mathematics of Shading / solid angle	Slower than PBRT with the algebra written out. The best free explanation of solid angle if the PBRT version moves too fast.
TU Wien Rendering — early lectures on radiometry	The intuition for why radiance is the transported quantity, developed visually.
PBRT 4th ed. — §5.4, The Camera Measurement Equation	The pixel integral of §5, derived properly, including the importance function that makes bidirectional methods possible.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Integrate the constant function 1 over the hemisphere numerically, in spherical coordinates. Confirm you get 2π . Then delete the $\sin\theta$ and report what you get instead — that number is the error you are protecting yourself against for the rest of the course.
- 2 Write the hemisphere check as a unit test in the repository you will use all semester. It should run in under a millisecond.
- 3 Compute the solid angle subtended by the sun from Earth, and by a 1 m square light panel at 3 m. Express both in steradians and as a fraction of the sphere.
- 4 Derive the radiance invariance result of §3 yourself, writing out both directions explicitly. Do not read it again first.
- 5 Verify the Lambertian normalisation: show that a BRDF of ρ/π conserves energy for $\rho \leq 1$, and explain where the π comes from. Most people get this wrong the first time.
- 6 Take a shading model from your CSCE 641 renderer and test it for reciprocity numerically over 1000 random direction pairs. Report the maximum relative asymmetry.
- 7 Test the same model for energy conservation: integrate $f \cos\theta$ over the hemisphere for twenty outgoing directions, including grazing ones. Report the maximum. Many popular models exceed 1 at grazing angles.
- 8 Look up the luminous efficacy of a 60 W-equivalent LED bulb and convert its lumen rating into watts of radiant flux, stating your assumption about its spectrum.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name the five radiometric quantities with their units, and say what distinguishes radiance from irradiance.
- 2 Give four reasons radiance is the quantity a renderer transports.
- 3 Write the solid angle element in spherical coordinates and say what goes wrong if the $\sin\theta$ is dropped.
- 4 Explain the cosine law in one sentence, without using the word 'Lambertian'. Then define what a Lambertian surface actually is.
- 5 Why is radiance invariant along a ray, and why does irradiance obey an inverse square law when radiance does not?
- 6 What does radiance invariance license, and which assumption does Module 10 remove?
- 7 Why is a BRDF not bounded by 1, and what quantity *is* bounded?
- 8 State the three required properties of a BRDF, and say which one causes divergence rather than merely incorrect output.

- 9 List the five dimensions of the pixel integral and the rendering effect each one produces.